E-Commerce Sales Analytics using SQL

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Course:

Data Analytics Project

Tools Used:

• SQL

• MySQL Workbench

• Power BI

Project Type:

SQL Data Analysis Project

Year:

2026

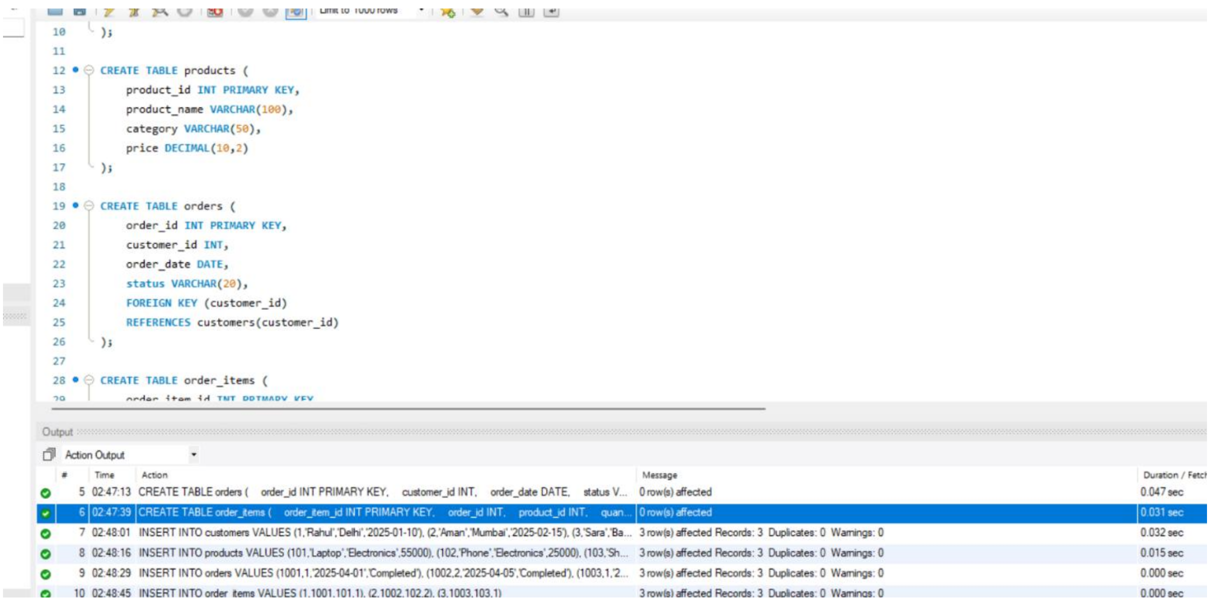
OBJECTIVE:-

The objective of this project is to analyze sales data, customer behavior, and product performance using SQL queries and data analytics techniques.

This project helps in understanding:

- Revenue analysis
- Top selling products
- Customer insights
- SQL joins and window functions
- Business decision making

SQL CREATE TABLE Queries:-



SQL INSERT Queries:-

```sql
);

INSERT INTO customers VALUES
(1,'Rahul','Delhi','2025-01-10'),
(2,'Aman','Mumbai','2025-02-15'),
(3,'Sara','Bangalore','2025-03-01');


INSERT INTO products VALUES
(101,'Laptop','Electronics',55000),
(102,'Phone','Electronics',25000),
(103,'Shoes','Fashion',3000);

INSERT INTO orders VALUES
(1001,1,'2025-04-01','Completed'),
(1002,2,'2025-04-05','Completed'),
(1003,1,'2025-04-10','Cancelled');

INSERT INTO order_items VALUES
(1,1001,101,1)
```

Output

Action Output

	#	Time	Action	Message
✔	5	02:47:13	CREATE TABLE orders (order_id INT PRIMARY KEY, customer_id INT, order_date DATE, status V...	0 row(s) affected
✔	6	02:47:39	CREATE TABLE order_items (order_item_id INT PRIMARY KEY, order_id INT, product_id INT, quan...	0 row(s) affected
✔	7	02:48:01	INSERT INTO customers VALUES (1,'Rahul','Delhi','2025-01-10'), (2,'Aman','Mumbai','2025-02-15'), (3,'Sara','Ba...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0

Total revenue :-

Limit to 1000 rows

```sql

SELECT
    SUM(p.price * oi.quantity) AS total_revenue
FROM order_items oi
JOIN products p
ON oi.product_id = p.product_id;

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

total_revenue
108000.00

Top Selling Products:-

```sql
SELECT
    p.product_name,
    SUM(oi.quantity) AS total_quantity
FROM products p
JOIN order_items oi
ON p.product_id = oi.product_id
```

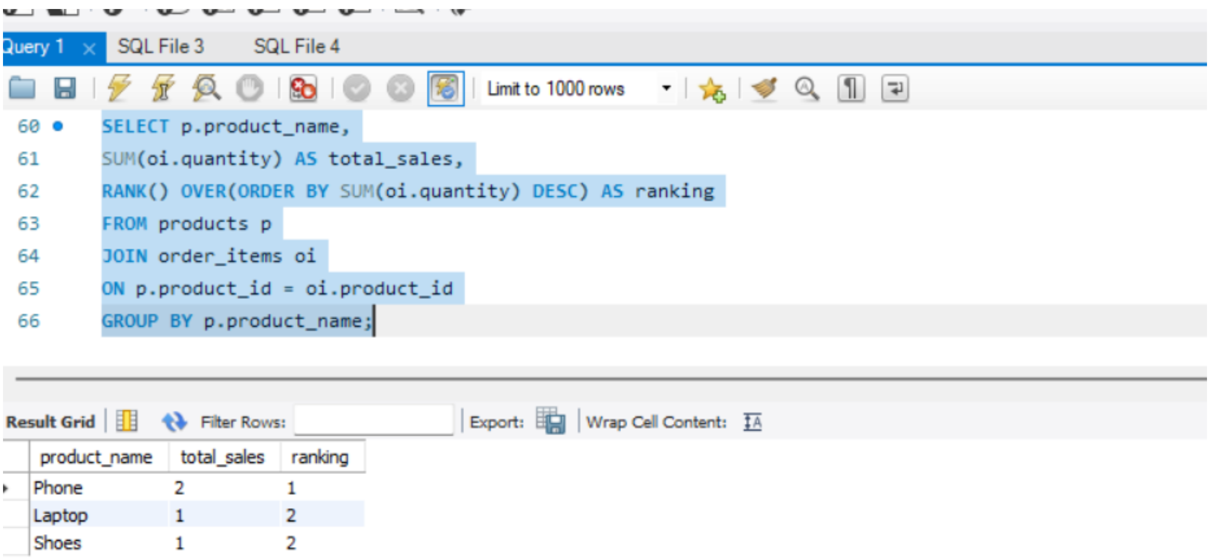
product_name	total_quantity
Phone	2
Laptop	1
Shoes	1

Top Customers by Revenue:-

```sql
SELECT c.customer_name, SUM(p.price * oi.quantity) AS total_spent
FROM customers c
JOIN orders o ON c.customer_id = o.customer_id
JOIN order_items oi ON o.order_id = oi.order_id
JOIN products p ON oi.product_id = p.product_id
GROUP BY c.customer_name
ORDER BY total_spent DESC;
```

customer_name	total_spent
Rahul	58000.00
Aman	50000.00

Product Sales Ranking using Window Function:-



Conclusion:-

Conclusion Paragraph

This project demonstrates the use of SQL for analyzing sales data, customer behavior, and product performance. Various SQL concepts such as JOIN, GROUP BY, aggregate functions, and window functions were used to generate business insights. This project helped in improving practical knowledge of SQL and data analytics.

Skills Learned

- SQL Queries
- Table Creation
- Data Insertion
- JOIN Operations
- Aggregate Functions
- Window Functions
- Data Analysis

Thank You